BCSE1563: DATA SCIENCE AND VISUALIZATION

Course Objectives: *This course introduces students to data science and visualization in the field of exploratory data science using Python.*

Prerequisites: Computer Programming - I

Credits: 03 **Semester - III** **L–T–P: 3–0–0**

Module No.	Content	Teaching Hours
I	**Introduction to Data Science**: Fundamentals of Data Science, Roles and responsibilities of a Data Scientist, Data Scientists Toolbox, Data Science Challenges, Exploratory Data Analysis and Data Science Process. Data and Data Models , Data Science methodology, Business understanding, Data Requirements, Data Acquisition, Data Understanding, Data preparation, Modelling, Model Evaluation, Motivation for using Python for Data Analysis, Introduction of Python shell iPython and Jupyter Notebook. **Type of Data** : Structured, Unstructured and Semi-Structured **Type of Analytics**: Descriptive, Predictive and Prescriptive **Sampling Techniques**: Simple Random Sampling, Systematic Sampling, Stratified Sampling, Cluster Sampling, Convenience Sampling, Non response , voluntary bias **Measurement & scaling**: Nominal, ordinal, interval, ratio **Statistical Analysis**- Central tendency, variance and Standard deviation, measure of dispersion (IQR range) , foundation of probability using python (Binomial, Normal distribution) **Built-In Data Structures, Functions, and Files:** Data Structures and Sequences, Functions, Files and the Operating System **Essential Python Libraries:** NumPy, pandas, matplotlib, SciPy, scikit-learn, statsmodels. **NumPy Basics: Arrays and Vectorized Computation:** The NumPy ndarray: A Multidimensional Array Object, universal functions, array operations. **Getting Started with Pandas**: Arrays and vectorized conputation, Introduction to pandas Data Structures, Essential Functionality, Summarizing and Computing Descriptive Statistics. **Data Cleaning and Preparation:** Handling Missing Data, Data Transformation, String Manipulation **Categorical Variables**: Creating Categorical Series, group by, setting order , adding and removing categories.	20
II	**Data Loading, Storage and File Formats:** Reading and Writing Data in Text Format, Web Scraping, Binary Data Formats, Interacting with Web APIs, Interacting with Databases **Data Wrangling:** Hierarchical Indexing, Combining and Merging Data Sets Reshaping and Pivoting. **Data Visualization matplotlib:** Basics of matplotlib, plotting with pandas and seaborn, other python visualization tools. **Data Aggregation and Group operations**: Group by Mechanics, Data aggregation, General split-apply-combine, Pivot tables and cross tabulation **Time Series Data Analysis**: Date and Time Data Types and Tools, Time series Basics, date Ranges, Frequencies and Shifting, Time Zone Handling, Periods and Periods Arithmetic, Resampling and Frequency conversion, Moving Window Functions. **Intro to ML:** Simple Linear Regression.	20

Text Books:

- McKinney, W.(2017). Python for Data Analysis: Data Wrangling with Pandas, NumPy and IPython. 2nd edition. OReilly Media
- ONeil, C., & Schutt, R. (2013). Doing Data Science: Straight Talk from the Frontline, OReilly Media
- VanderPlas, J. (2016). Python data science handbook: Essential tools for working with data. OReilly Media

Reference Books:

- Sweigart, A. (2019). Automate the boring stuff with Python: practical programming for total beginners. No Starch Press.
- Grus, J. (2019). Data science from scratch: first principles with python. O'Reilly Media.

Focus: This Course focuses on Employability under CO2, CO3, and CO4.

Program Name: B.Tech. CSE

Department: Department of Computer Engineering & Applications

Outcome: The student will be able to

CO 1. Use data analysis tools in the pandas library
CO 2. Load, clean, transform, merge and reshape data.
CO 3. Handle external files as well as exceptions.

CO 4. Analyze and manipulate time series data.
CO 5. Solve real world data analysis problems

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	**POs/PSOs**
CO1	PO1/PSO3
CO2	PO3/PSO4
CO3	PO2,PO3,PO11/PSO2,PSO4
CO4	PO5,PO2/PSO4
CO5	PO2,PO3/PSO2